

DLGIN: Deep Learning Genre Identification Network

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Agenda

What is DLGIN

Why DLGIN

How it works & how we made it

Demo

What's next

Questions

DLGIN:

Deep Learning Genre Identification Network

A statistical analysis tool that uses machine learning (ML) and music information retrieval (MIR) techniques to identify music as a combination of 16 genres



Why DLGIN?

Who we are:

Sean: Computer Science + Math
Justin: Computer Science + Political Science
Matt: Computer Science + Military Science



Why DLGIN?



















Why DLGIN?

- A system like this doesn't really exist yet
- Music streaming platforms have a vested interest in developing music technology
- Would allow us to explore (and maybe even contribute to) the field of music information retrieval





How It Works

Music Source: Free Music Archive

Web Framework: Django

ML Libraries: Scikit-Learn & Keras

MIR Library: Librosa

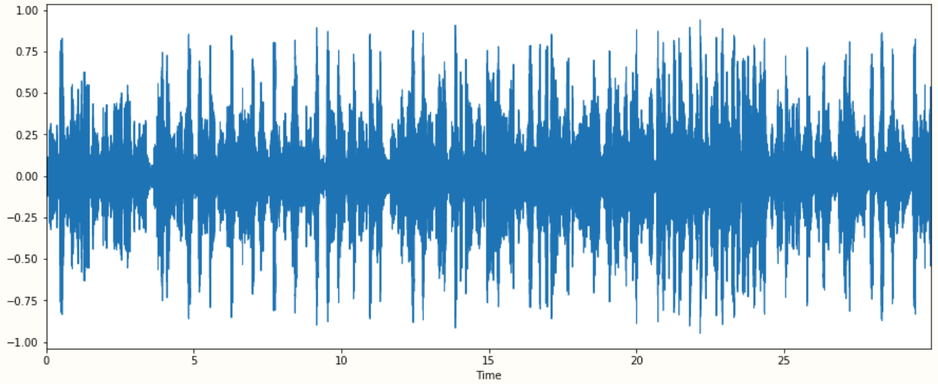


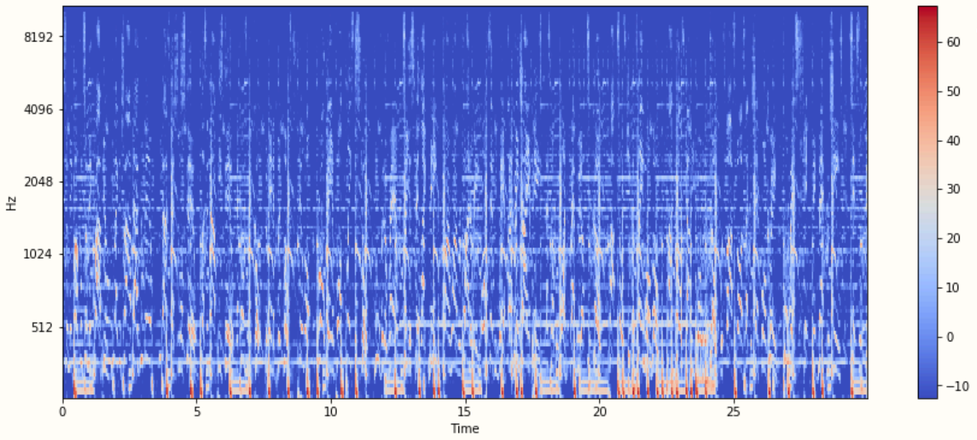
How It Works

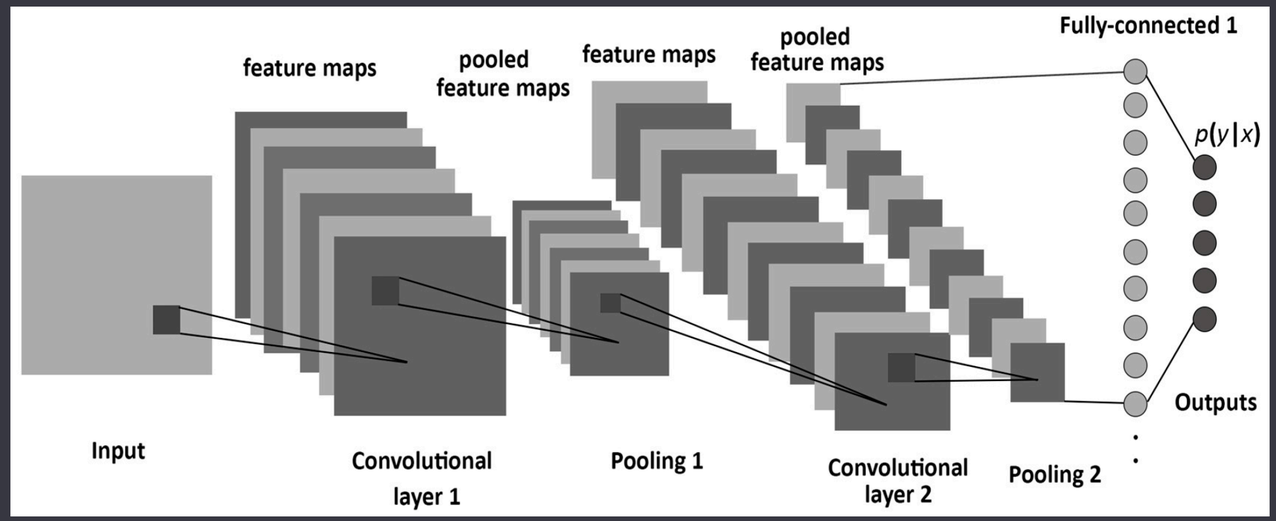
Models Tested:

1. Multi-layer Perceptron Classifier (MLPC)
2. Gradient Boosting Classifier (GBC)
3. Ada Boosting Classifier (ABC)
4. Multiple Linear Regression (LIN)
5. Multiple Logistic Regression (LOG)
6. Convolutional Neural Network (CNN)
7.) Convolutional Recurrent Neural Network (CRNN)

How It Works

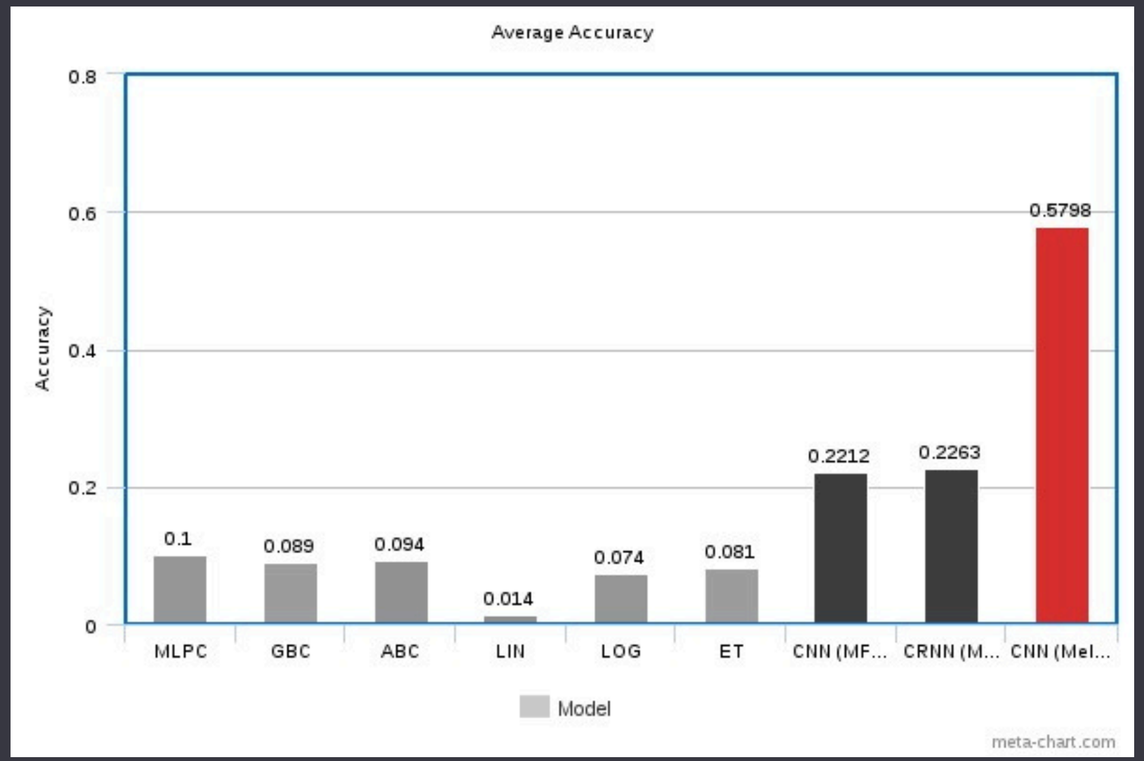

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25
Time


8192
4096
2048
1024
512
Hz
0
5
10
15
20
25
Time
60
50
40
30
20
10
0
−10


feature maps
pooled feature maps
feature maps
pooled feature maps
Fully-connected 1
p(y | x)
Outputs
Input
Convolutional layer 1
Pooling 1
Convolutional layer 2
Pooling 2



How It Works


Average Accuracy
Accuracy
0.8
0.6
0.4
0.2
0
0.1
0.089
0.094
0.014
0.074
0.081
0.2212
0.2263
0.5798
MLPC
GBC
ABC
LIN
LOG
ET
CNN (MF...
CRNN (M...
CNN (Mel...
Model
meta-chart.com



Demo!



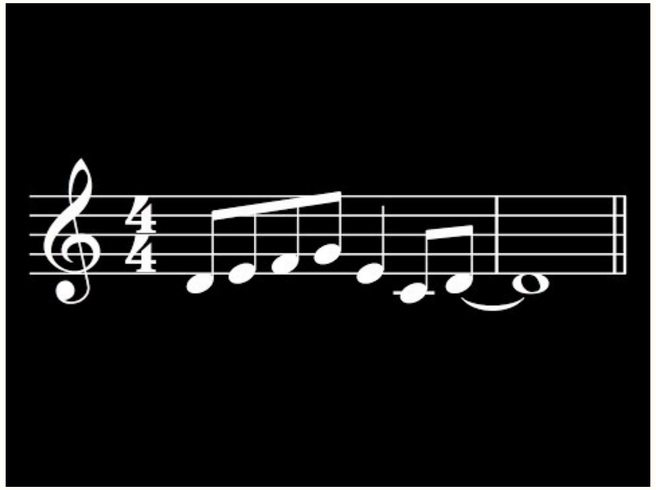
Next Steps

DLGIN 2.0:

Improve accuracy!

Integration with Spotify API

Recommendation system



(Thank You)